

1 P3 C3 — Nontriangular Bulgarian solitaire: bound and all maximizers

Exact cell statement. This is a written general proof. The accompanying Lean theorems certify finite ranks only; the finite replay below corroborates, but does not replace, the all- k argument.

A partition $\lambda = (\lambda_1, \dots, \lambda_s)$ represents n cards in positive piles. The Bulgarian move B subtracts one card from each pile, discards zero piles, adds a new pile of size s , and sorts. A state is *cyclic* if $B^a(\lambda) = \lambda$ for some $a > 0$. Let $d_B(\lambda)$ be the least $t \geq 0$ for which $B^t(\lambda)$ is cyclic, and let $D_B(n) = \max_{\lambda \vdash n} d_B(\lambda)$. Set $T_j = j(j+1)/2$.

We use the established **C1 cyclic-boundary theorem**: at rank k , where $n = T_{k-1} + r$ and $1 \leq r \leq k$, the cyclic partitions are precisely the decreasing lists $k-1+\varepsilon_0, k-2+\varepsilon_1, \dots, \varepsilon_{k-1}$, omitting a final zero, with $\varepsilon_j \in \{0, 1\}$ and $\sum \varepsilon_j = r$. Their boundary words rotate under B , so their periods divide k ; each has pile sizes at most k , and for $1 \leq r < k$ the pile-count cycle includes a rise from $k-1$ to k . The cited C1 proof also establishes eventual cycling.

1.1 Full C3 proof: nontriangular bound and extremal size

Let $T_j = j(j+1)/2$, and write every nontriangular card count as

$$n = T_{k-1} + r, \quad 1 \leq r \leq k-1.$$

For $k \geq 4$, the maximum first-cyclic depth satisfies

$$D_B(n) \leq k^2 - 2k - 1.$$

(A)

At $n = T_k - 1$, equality holds. The section below also gives an exact, computable criterion for **every** maximizing starting partition at that size. This is a written proof; the general C3 statements have not been formalized in Lean. The existing Lean files verify the finite rank-4-through-rank-6 values. The published source is [Griggs and Ho, Theorem 4.4](#); the argument here supplies the pile-lifetime details behind its upper bound and an explicit trajectory-count classification of equality.

1.1.1 Sequence facts carried over from C2

For a starting partition λ , let c_i be the number of piles in $B^{i-1}(\lambda)$. The new pile born on move i has size c_i and lives at times $i+1, \dots, i+c_i$. Denote its lifetime interval by $J_i = [i+1, i+c_i]$. At time m , exactly c_m intervals, including those of the original piles, are alive. Hence $c_{i+1} \leq c_i + 1$; a rise by one means no pile dies on move i .

The **C2 upper-bound proof** proves three sequence lemmas without using the C2 conclusion:

1. If $c_i \leq x-1$ and $c_j \geq x+1$ for $i < j$, a *sandwich pattern* $(c_p, \dots, c_q) = (x-1, x, \dots, x, x+1)$ occurs inside $[i, j]$.
2. Every such pattern of width $L = q - p \geq 2$ obeys $p \leq x(L-1)$. This follows from the explicit retreat to a shorter earlier pattern, stopping when $p \leq x$, as is forced at width two.

3. If $(c_p, \dots, c_{p+k}) = (k-2, k-1, \dots, k-1, k)$, then $p+k \leq n+1$. The proof counts live intervals at time $p+k$ and rules out an extra pile born after move 1.

We also need two elementary links between a pile-count sequence and its state. First, its entire future determines its starting partition. At relative time m , subtract from c_m the number of known new intervals J_i covering m . The result is the number of original piles of size at least m , for every m , and these counts determine the partition. Thus, if $c_{t+i} = c_{t+k+i}$ for all $i \geq 0$, the states at times t and $t+k$ are equal; the state at time t is cyclic.

Second, let $m \geq 1$ satisfy

$$c_m = k-1, \quad c_{m+1} = k, \quad c_{m+a} \in \{k-1, k\} \quad (2 \leq a < k), \quad c_{m+k} = k-1.$$

(B)

The state at time m is already cyclic. Indeed, for $1 \leq a \leq k$, all the $a-1$ piles born on moves $m, \dots, m+a-2$ remain alive at time $m+a-1$. If H_a counts the piles of the time- m state that have size at least a , then

$$H_a = c_{m+a-1} - (a-1) \in \{k-a, k-a+1\}.$$

(C)

At time $m+k$, the $k-1$ piles $J_{m+1}, \dots, J_{m+k-1}$ already fill all $c_{m+k} = k-1$ places, so no time- m pile has size above k . Formula (C) says the conjugate diagram lies between the staircases of sizes $k-1$ and k ; transposing gives the same containment for the original diagram. Its column heights therefore have the form $k-1 + \varepsilon_0, k-2 + \varepsilon_1, \dots, \varepsilon_{k-1}$, with bits ε_j , so it is cyclic by the [C1 classification](#).

1.1.2 Finding a late sandwich

The C1 boundary classification also says every cyclic state at rank k has pile counts in $\{k-1, k\}$ and period dividing k . Because $1 \leq r < k$, its boundary word contains both a zero and a one, so the cyclic count sequence has a rise $k-1$ to k . Choose the **smallest** positive index t for which

$$c_t = k-1, \quad c_{t+1} = k, \quad c_{t+i} = c_{t+k+i} \quad \text{for all } i \geq 0.$$

(D)

This t exists. Let $d_B(\lambda)$ be the first-cyclic depth. The future-count argument above gives $d_B(\lambda) \leq t-1$. If $t \geq k+1$, minimality means the preceding block $(c_{t-k}, \dots, c_{t-1})$ differs from $(c_t, \dots, c_{t+k-1})$; otherwise the periodicity and the rise in (D) would already begin at $t-k$.

Assume $t \geq k+1$. Since $c_t = k-1$, one of the k recent intervals $J_{t-k}, \dots, J_{t-1}$ is absent at time t . Let u be the largest absent index. The rise $c_t \rightarrow c_{t+1}$ entails no death, so the same lifetime calculation as in C2 gives

$$c_u = t - u - 1, \quad c_{u+1} = t - u.$$

(E)

If $u \geq t - k + 1$, then $c_u \leq k - 2$, and the sandwich rule between u and $t + 1$ produces a **type II** pattern

$$(c_p, \dots, c_q) = (k - 2, k - 1, \dots, k - 1, k), \quad t - k + 1 \leq p < q \leq t + 1, \quad q - p \leq k.$$

(II)

Otherwise $u = t - k$, so $c_{t-k} = k - 1$ and $c_{t-k+1} = k$. If one of $c_{t-k+2}, \dots, c_{t-1}$ is at most $k - 2$, the same rule gives (II). If one is at least $k + 1$, the rule between $t - k$ and that index produces a **type I** pattern

$$(c_p, \dots, c_q) = (k - 1, k, \dots, k, k + 1), \quad t - k \leq p < q \leq t - 1, \quad q - p \leq k - 1.$$

(I)

In the remaining case all intermediate counts lie in $\{k - 1, k\}$. Since (D) also gives $c_t = k - 1$, condition (B) holds at $m = t - k$, so that earlier state is cyclic. Its future counts must then be k -periodic, contrary to the minimality of t . Thus (I) or (II) always occurs.

1.1.3 The upper bound

For $k = 4$, the exact [Lean checks for \$n = 7, 8, 9\$](#) give depths 4, 5, 7, all at most $4^2 - 2 \cdot 4 - 1 = 7$. Let $k \geq 5$, and put $B = k^2 - 2k - 1$. If $t \leq k$, then $d_B(\lambda) \leq t - 1 \leq k - 1 < B$. Otherwise take the pattern just found.

For type II of full width k , its index range forces $(p, q) = (t - k + 1, t + 1)$. The special sequence lemma gives

$$d_B(\lambda) \leq t - 1 = p + k - 2 \leq n - 1 < B.$$

The last inequality follows from $n \leq T_k - 1$ and $T_k - 2 < B$ for $k \geq 5$.

Width $k - 1$ is impossible for either type. For type I, the index range forces $q = t - 1$, so $c_{t-1} = k + 1$. Move $t - 1$ then creates a pile of size $k + 1$ in the cyclic state at time t , impossible because every rank- k cyclic pile has size at most k . For type II, the index range and $c_t = k - 1$ force $(p, q) = (t - k + 2, t + 1)$: the only other possible placement would have $q = t$ and $c_t = k$. But $c_p = k - 2$ makes J_p end at $p + k - 2 = t$, whereas $c_t = k - 1 \rightarrow c_{t+1} = k$ permits no pile to end at t .

Every remaining pattern has width $L \leq k - 2$, level $x \in \{k - 1, k\}$, and $p \geq t - k$. The C2 retreat bound yields

$$d_B(\lambda) \leq t - 1 \leq p + k - 1 \leq x(L - 1) + k - 1 \leq k(k - 3) + k - 1 = B.$$

This proves (A).

1.1.4 Equality at $T_k - 1$

For every $k \geq 4$, begin with the staircase $(k, k-1, \dots, 1)$, delete its two cells on diagonal k in columns 0, 1, and add one cell on diagonal $k+1$ in column k . The resulting partition of $T_k - 1$ is

$$\lambda^* = (k-1, k-2, k-2, k-3, \dots, 1, 1).$$

(F)

This is the witness of [Griggs and Ho, Theorem 4.4\(2\)](#). Put $B = k^2 - 2k - 1$. For $0 \leq t < B$, define a diagram D_t by deleting the diagonal- k cells in columns

$$p_t = t \bmod k, \quad p'_t = (p_t + 1) \bmod k,$$

and adding the diagonal- $(k+1)$ cell in column $q_t = (k+t) \bmod (k+1)$. Here a cell on diagonal w in column j has row $w - j$. We check that every D_t is a Ferrers partition. If $p_t \leq k-2$, the only invalid added-column positions are $q_t \in \{p_t, p_t+1, p_t+2\}$: they create a vertical gap or reverse adjacent heights. If $p_t = k-1$, the holes wrap to columns $k-1, 0$, and the invalid positions are $q_t \in \{k-1, k, 0, 1\}$.

Write $t = ak + b$, $0 \leq b < k$. For $t < B$, we have $a \leq k-3$, excluding $(a, b) = (k-3, k-1)$, and

$$q_t = \begin{cases} b - a - 1, & b \geq a + 1, \\ k + b - a, & b \leq a. \end{cases}$$

(G)

If $b \leq k-2$, the first branch gives $q_t < b$, while the second gives $q_t \geq b+3$, so all three forbidden positions are avoided. If $b = k-1$, then $a \leq k-4$ and $q_t = k - a - 2 \in \{2, \dots, k-2\}$, avoiding all four wrapped forbidden positions. Thus every D_t before B is Ferrers. Before sorting, a Bulgarian move rotates each diagonal's column coordinates by one, so $B(D_t) = D_{t+1}$ whenever $t+1 < B$. Since $D_0 = \lambda^*$, induction gives $B^t(\lambda^*) = D_t$ for all $t < B$.

For the Ferrers check above, when the added column differs from both deleted columns, its heights are explicitly $h_j = k - j - \mathbf{1}_{j=p_t} - \mathbf{1}_{j=p'_t} + \mathbf{1}_{j=q_t}$ for $0 \leq j \leq k$. Adding at a deleted column leaves a vertical gap. In all other cases, inspecting adjacent differences $h_j - h_{j+1}$ gives exactly the forbidden positions stated above; no other pair of heights reverses.

At $t = B - 1$, the holes are in columns $k-2, k-1$ and the added cell is in column 0, giving

$$B^{B-1}(\lambda^*) = (k+1, k-1, k-2, \dots, 3, 1).$$

Its next unsorted move has column heights $(k-1, k, k-2, \dots, 2)$; sorting the first two yields $(k, k-1, k-2, \dots, 2)$, a rank- k cyclic boundary partition of $T_k - 1$. Every earlier D_t has an occupied cell on diagonal $k+1$, while a rank- k cyclic partition has none, by C1. Hence $d_B(\lambda^*) = B$, and the upper bound proves $D_B(T_k - 1) = B$.

1.1.5 Which starting partitions attain the maximum?

There is an exact pile-count signature for all maximizers. For $k \geq 5$, set $n = T_k - 1$, $B = k^2 - 2k - 1$, and $P = k(k - 3)$. A partition $\lambda \vdash n$ has depth B **if and only if**

$$(c_P, c_{P+1}, \dots, c_{P+k-2}) = (k - 1, k, \dots, k, k + 1).$$

(H)

The right side has $k - 1$ terms: one $k - 1$, then $k - 3$ copies of k , then one $k + 1$. For necessity, equality in the upper-bound chain forces type I (type II has $t \leq p + k - 1$), level $x = k$, width $L = k - 2$, and start $p = k(k - 3)$, giving (H). For sufficiency, its last term is $c_{B-1} = k + 1$. Move $B - 1$ creates a pile of size $k + 1$ in $B^{B-1}(\lambda)$, so that state is not cyclic. Once a trajectory enters a cycle it never leaves; hence $d_B(\lambda) \geq B$, and (A) gives equality. For $k = 4$, direct enumeration of the 30 partitions of 9 gives depth counts 4, 3, 6, 7, 4, 3, 2, 1 at depths 0, 1, $\dots$, 7, respectively. The unique depth-seven partition is (3, 2, 2, 1, 1), which also satisfies (H) with $P = 4$. The exact Lean certificate verifies the maximum depth; the identifying enumeration can be replayed from recurrence (I) and the four cyclic boundary partitions.

The signature in fact fixes a particular late state. Let $q = P + k - 2 = B - 1$. At time q , the piles born on moves $P, \dots, q - 1$ are all alive. The pile J_P , born with size $k - 1$, now has size 2; those born on moves $P + 1, \dots, q - 1$, each with initial size k , have sizes 4, 5, $\dots$, k . These are $k - 2$ piles containing $2 + 4 + 5 + \dots + k = T_k - 4$ cards. Since $c_q = k + 1$ and $n = T_k - 1$, the other three piles have three cards in total, so each has size 1. Therefore every maximizer satisfies

$$B^{B-2}(\lambda) = Q_k := (k, k - 1, \dots, 4, 2, 1, 1, 1).$$

(J)

Conversely, Q_k has depth exactly two. It has $k + 1$ piles, so its first move produces

$$P_k = (k + 1, k - 1, k - 2, \dots, 3, 1),$$

whose next move is the cyclic boundary $(k, k - 1, \dots, 2)$. Both Q_k and P_k are noncyclic: the former has too many piles for a rank- k cyclic partition, and the latter has a pile larger than k . Thus (J) is another necessary and sufficient condition for depth B .

Equation (J) gives a finite **inverse construction of all starting maximizers**, using only a fixed seed and an explicit rule on partitions. For a partition μ with m parts, choose any distinct part value s of μ such that $s \geq m - 1$. Remove one occurrence of s , add 1 to each remaining part, append $s - m + 1$ parts of size 1, and sort; call the result $R_s(\mu)$. It has exactly s parts and $B(R_s(\mu)) = \mu$. Conversely every predecessor of μ arises this way, because its new pile must have size s , one of the parts of μ .

Start with $\mathcal{S}_0 = \{Q_k\}$ and repeat

$$\mathcal{S}_{j+1} = \bigcup_{\mu \in \mathcal{S}_j} \{R_s(\mu) : s \text{ is a distinct part of } \mu, s \geq \ell(\mu) - 1\}, \quad 0 \leq j < B - 2.$$

(K)

Then $\mathcal{S}_{B-2}$ is **exactly** the set of all maximizing partitions of $T_k - 1$. Every state in that final set reaches Q_k after $B - 2$ moves, so has depth B ; every maximizer belongs to the set by (J) and the exhaustive predecessor rule. The construction never tests whether a state is cyclic and does not use a depth oracle. It can merge duplicate branches as sets. For $k = 4$, it yields the sole partition $(3, 2, 2, 1, 1)$; for larger k , the witness (F) is one member among many.

Criterion (H) is a compact membership test for the same set. Its counts can be computed directly from the initial parts by the finite recurrence

$$c_m = |\{j : \lambda_j \geq m\}| + |\{1 \leq i < m : i + c_i \geq m\}|, \quad m \geq 1.$$

(I)

The inverse construction (K) and count recurrence (I) are exact constructive answers to which partitions attain the maximum. They do not give a closed-form number of maximizers or a single static Ferrers-inequality description; neither is needed to generate or recognize every one.

1.1.6 Exhaustive finite replay

Run `python3 cagent/p3/c3/check.py` from the repository root. It independently computes forward depths and constructs the complete inverse sets for $k = 4, 5, 6, 7$, then compares the sets elementwise. [Recorded results](#) have respectively 1, 6, 34, and 175 maximizing partitions. This also supplies the finite $k = 4$ classification used above.

1.2 Appendix: pile-count lemmas used above

The next three sections reproduce the full general sequence arguments from the published [C2 upper-bound proof](#). C3 uses these lemmas, not the triangular C2 conclusion.

1.2.1 Pile lifetimes and a sequence bound

Write c_i for the number of piles in $B^{i-1}(\lambda)$, with $i \geq 1$. An original pile of size a exists at times $1, \dots, a$. The pile created by move i has size c_i and exists at times

$$J_i = [i + 1, i + c_i].$$

Its last time is $e_i = i + c_i$. At any time m , precisely c_m of these original-pile and created-pile intervals contain m . One new interval starts between times i and $i + 1$; if d_i intervals end at i , then

$$c_{i+1} = c_i + 1 - d_i.$$

(1)

In particular $c_{i+1} \leq c_i + 1$; a rise by one means **no** pile dies, and a constant pair means **exactly one** pile dies.

We use a *sandwich pattern* of level x and width $L = q - p \geq 2$:

$$(c_p, c_{p+1}, \dots, c_q) = (x - 1, x, \dots, x, x + 1).$$

(2)

The two rises occur at its ends. The following elementary sandwich rule will locate one. If $i < j$, $c_i \leq x - 1$, and $c_j \geq x + 1$, choose p as the last index before j with $c_p \leq x - 1$, and q as the first subsequent index with $c_q \geq x + 1$. Equation (1) forces $c_p = x - 1$, $c_{p+1} = \dots = c_{q-1} = x$, and $c_q = x + 1$. Thus (2) occurs within $[i, j]$.

1.2.2 The retreat lemma

Suppose (2) occurs and $p > x$. Among the x recent created piles $J_{p-x}, \dots, J_{p-1}$, at least one is absent at time p , since $c_p = x - 1$. The last, J_{p-1} , is present. Let $u \in [p - x, p - 2]$ be the largest index whose interval is absent. All of $J_{u+1}, \dots, J_{p-1}$ are present at p . Since $c_{p+1} = c_p + 1$, none dies at p , so they remain present at $p + 1$.

Put $h = p - u - 1$ and $y = h + 1 = p - u \leq x$. The absence of J_u at p gives $c_u \leq h$. The presence of J_{u+1} at $p + 1$ gives $c_{u+1} \geq h + 1$. Applying $c_{u+1} \leq c_u + 1$ shows

$$c_u = h = y - 1, \quad c_{u+1} = h + 1 = y.$$

(3)

Moreover J_{u+1} ends at time $u + 1 + y = p + 1$.

If $L = 2$, the pattern also has $c_{p+2} = c_{p+1} + 1$, so no interval ends at $p + 1$, a contradiction. Hence every width-two pattern satisfies

$$p \leq x.$$

(4)

Now let $L \geq 3$. As long as no later value c_{u+s} has risen from y to $y + 1$, the consecutive intervals

$$J_{u+s}, J_{u+s+1}, \dots, J_{p+s-1}$$

are all present at time $p + s$, for $1 \leq s \leq L - 1$. This holds for $s = 1$ by the choice of u and the lack of a death at p . If $c_{u+s} = y$, then J_{u+s} ends at $(u + s) + y = p + s$. For $s \leq L - 2$, the pair $c_{p+s} = c_{p+s+1} = x$ permits exactly one death. It must be this interval; all the others survive, and J_{p+s} is born. This proves the induction. At each stage the active J_{u+s} implies $c_{u+s} \geq y$, while (1) and the preceding value y imply $c_{u+s} \leq y + 1$. Hence it either stays at y or supplies the desired rise.

At $s = L - 1$, the transition from $c_{q-1} = x$ to $c_q = x + 1$ permits no death. Thus J_{u+L-1} cannot have size y , which would make it end at $q - 1$. A first rise occurs at some v with

$$u + 2 \leq v \leq u + L - 1, \quad (c_u, \dots, c_v) = (y - 1, y, \dots, y, y + 1).$$

(5)

It occurs by $p+1$: otherwise both c_p and c_{p+1} would equal y , contradicting their values $x-1, x$. Thus $u \geq p-x$, $v \leq p+1$, $y \leq x$, and the new width $v-u < L$. This is the retreat step.

Each retreat decreases the positive integer width. It therefore stops either at a pattern with start at most its level or at width two, where (4) gives the same bound. Working backward through $p \leq u+x$, while levels never increase, gives the explicit bound

$$\boxed{p \leq x(L-1)}$$

(6)

for every sandwich pattern (2). More formally, if $p \leq x$, then $p \leq x(L-1)$ since $L \geq 2$. Otherwise induction on L gives $p \leq u+x \leq y(L'-1) + x \leq x(L-2) + x = x(L-1)$ for the smaller width $L' < L$. This abstract induction is also kernel-checked as [sandwich_start_bound](#); the lifetime argument above proves its application hypotheses in writing.

1.2.3 A special full-width pattern

We also need one bound that uses the number n of cards. Suppose

$$(c_p, \dots, c_{p+k}) = (k-2, k-1, \dots, k-1, k).$$

(7)

Here $k \geq 3$. The interval J_p ends at $p+k-2$. Since $c_{p+k-2} = c_{p+k-1} = k-1$, it is the **only** interval ending then. No interval ends at $p+k-1$, since the next count rises from $k-1$ to k .

At time $p+k$, the $k-1$ intervals $J_{p+1}, \dots, J_{p+k-1}$ are all present. There is exactly one other pile. If it is original, its original size is at least $p+k$, and hence $n \geq p+k$. If it is J_1 , then $1+c_1 \geq p+k$, and $c_1 \leq n$ gives $n+1 \geq p+k$.

The only other possibility would be J_i for some $2 \leq i \leq p-1$; J_p has already died. Such a J_i is also the unique extra pile at time $p+k-1$, alongside $J_{p+1}, \dots, J_{p+k-2}$. Hence J_{i-1} is absent at that time, so $(i-1) + c_{i-1} \leq p+k-2$. But J_i survives to $p+k$, and (1) yields

$$(i-1) + c_{i-1} \geq (i-1) + (c_i - 1) \geq p+k-2.$$

Equality follows, making J_{i-1} a second interval ending at $p+k-2$, contrary to the uniqueness of J_p . We have proved

$$\boxed{p+k \leq n+1}.$$

(8)